

Dynamic Exogenous Context for Implicit Neural Network

A Continuous Alternative to LSTMs in Time Series Forecasting

Introducing Myself

PhD student in Artificial Intelligence & Data Science

Université Paris-Saclay · MIA Paris-Saclay · EKINOCS team

PhD thesis sujet

Modeling and Prediction of Contextualized Time Series

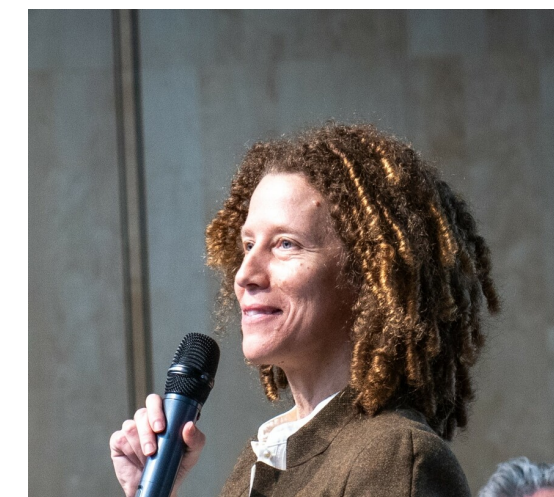


Jessy Azizi

PhD supervisor:



Vincent Guigue



Sophie Martin

A Continuous Alternative to LSTMs in Time Series Forecasting

Introducing Myself

Research Focus

My PhD focuses on developing deep learning and hybrid approaches for **time series forecasting**, combining data-driven methods with mechanistic approaches.

Current focus: incorporating **contextual information** into time series models to improve forecasting performance and better capture the underlying dynamics.

A Continuous Alternative to LSTMs in Time Series Forecasting

What are we gonna talk about?

An implicit neural représentation architecture (INR)

- What are INRs ?
- How do they work ?
- What are their particularities ?

1/3

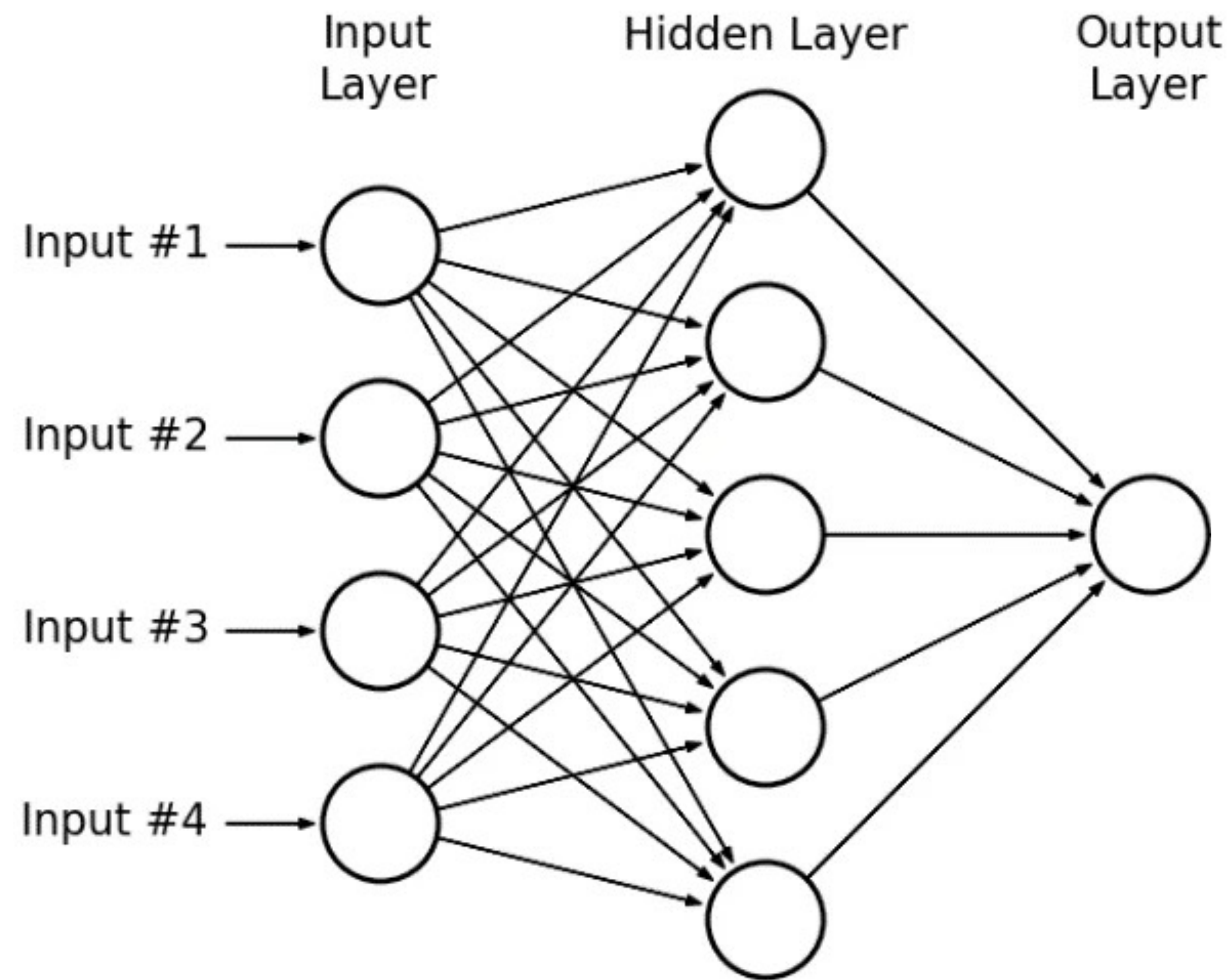
Our contribution on the implicit neural representation

2/3

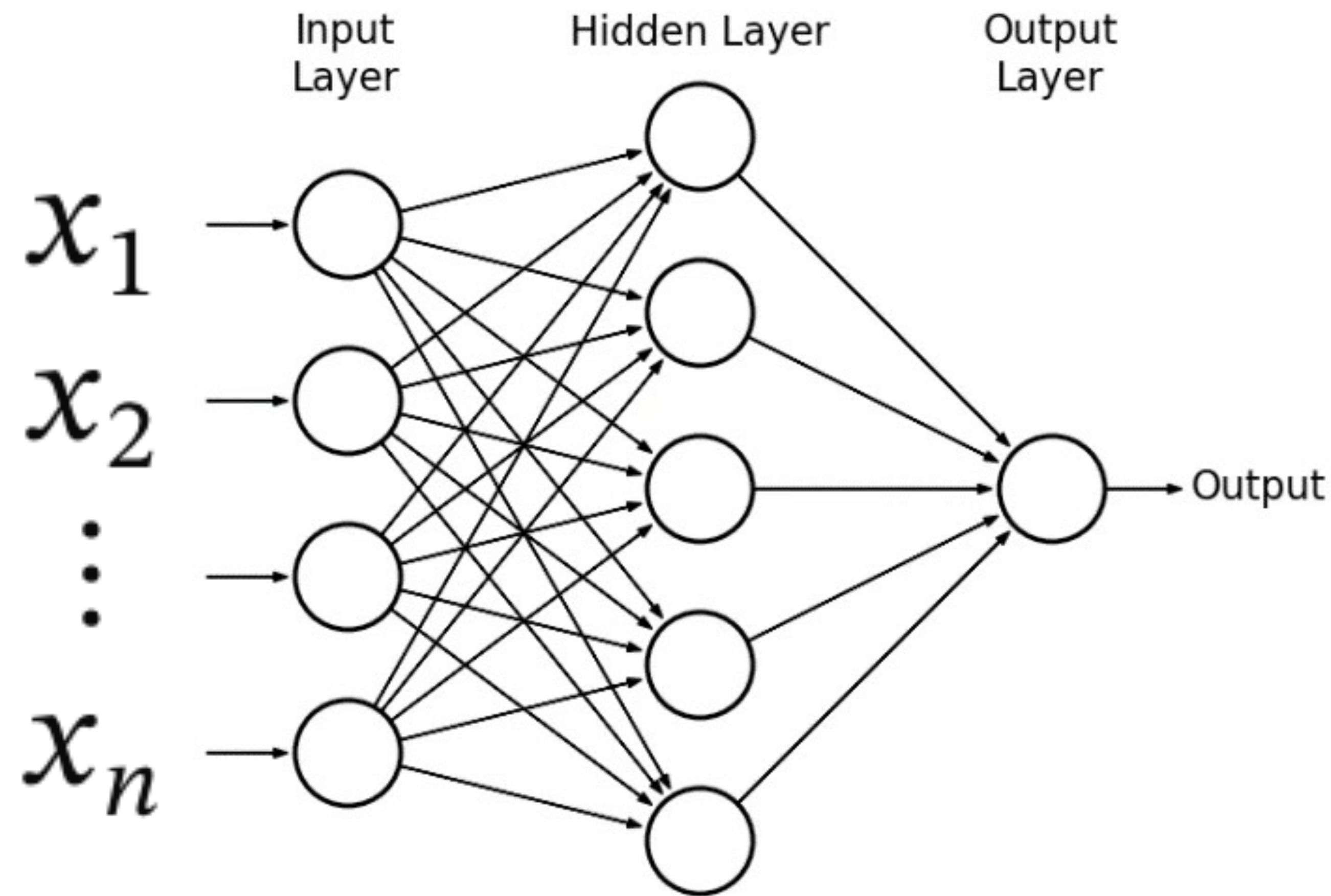
Apply implicit neural representation to electricity price forecasting data

3/3

Implicit neural representation

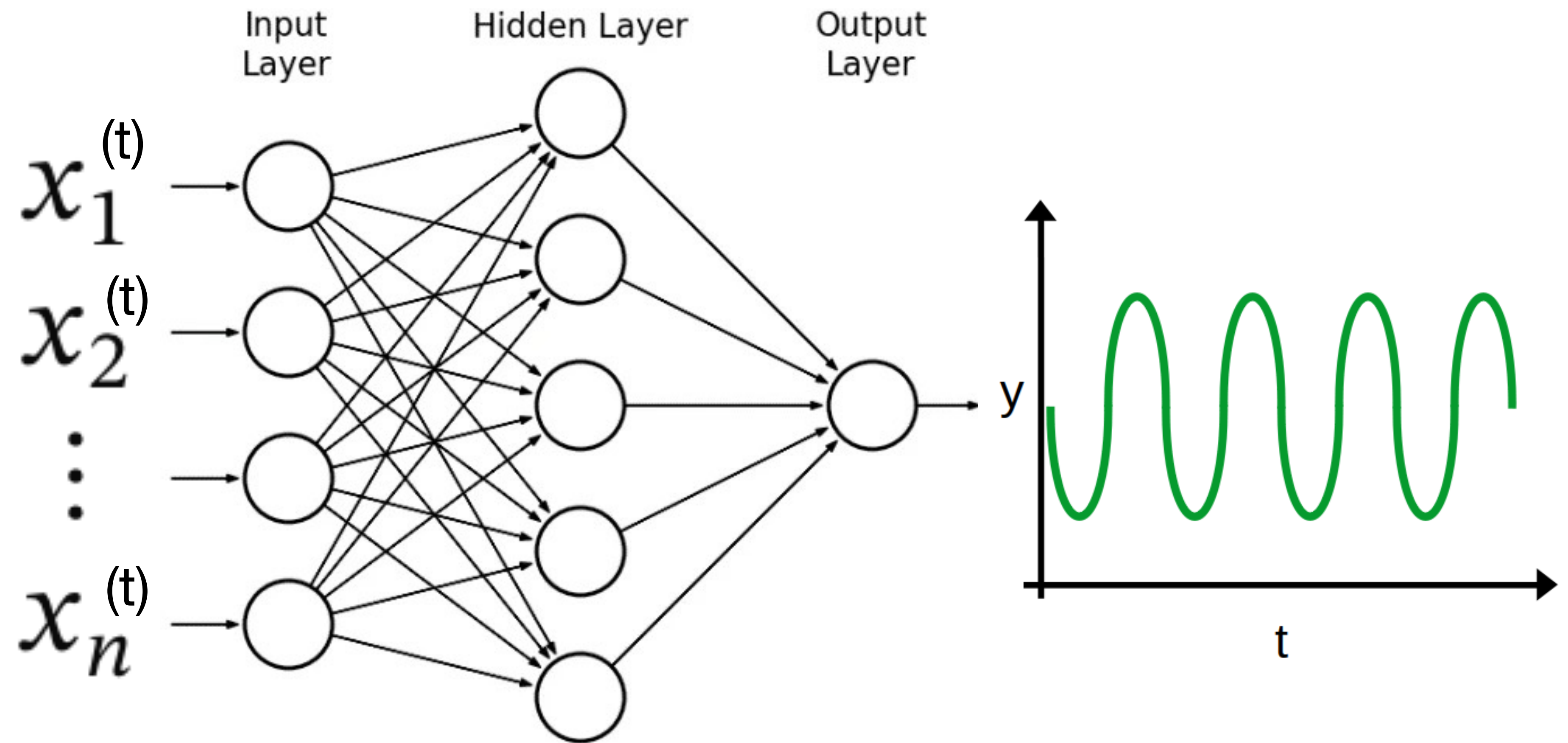


Implicit neural representation



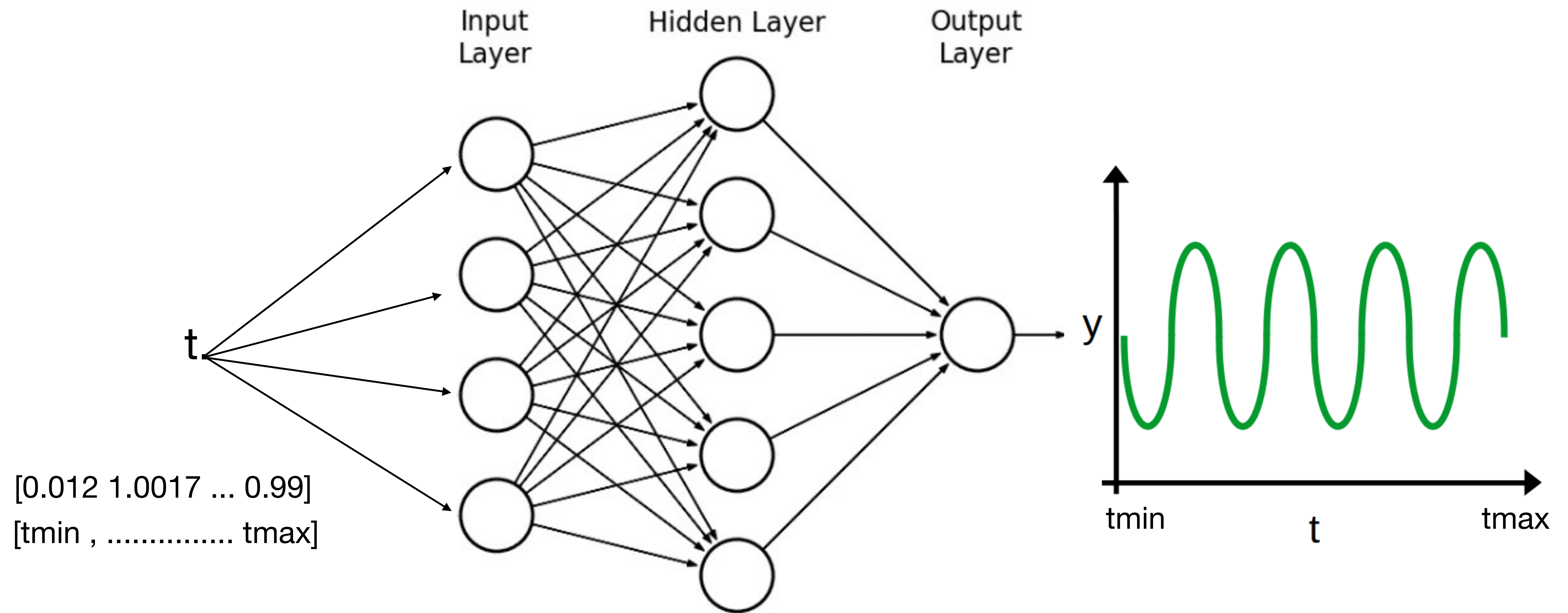
features vector

Implicit neural representation

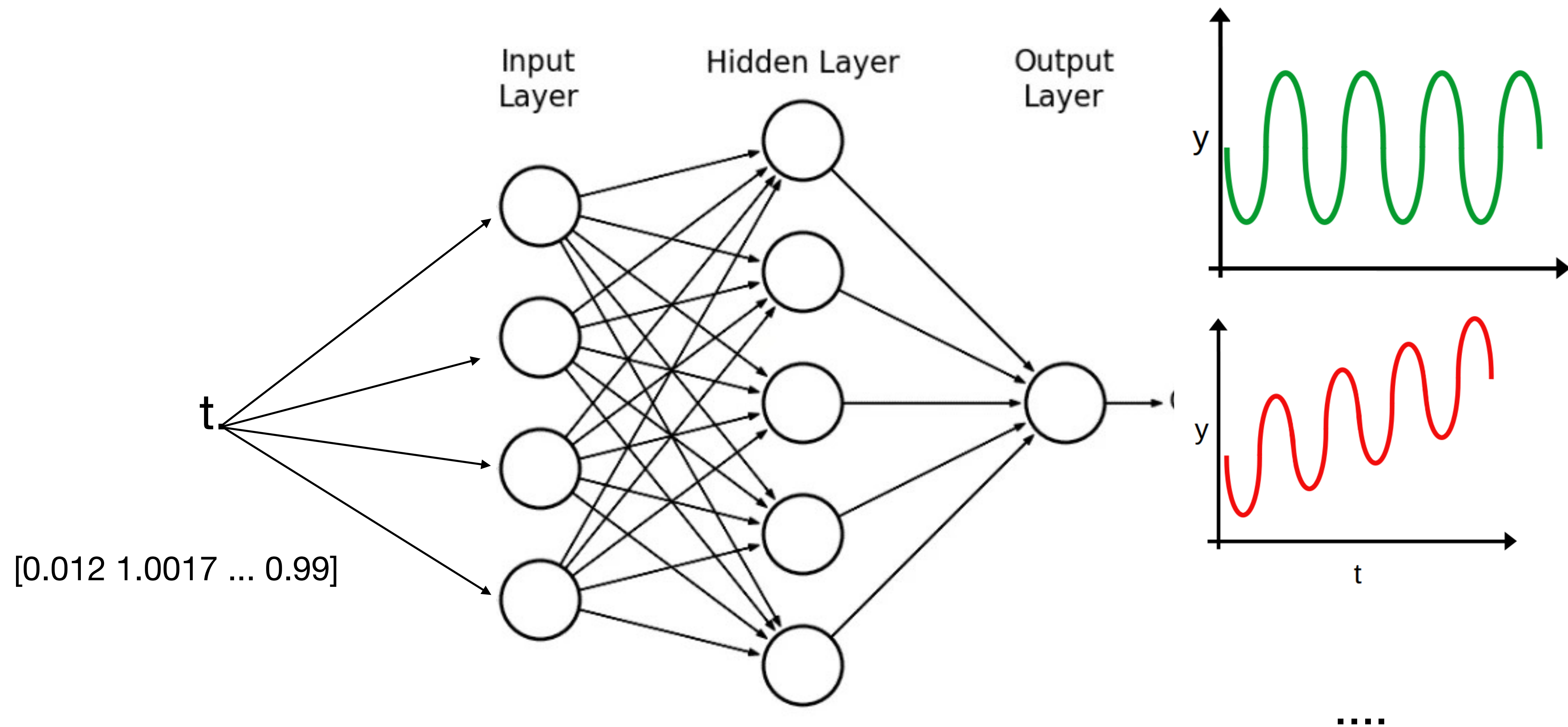


t * features vector

Implicit neural representation

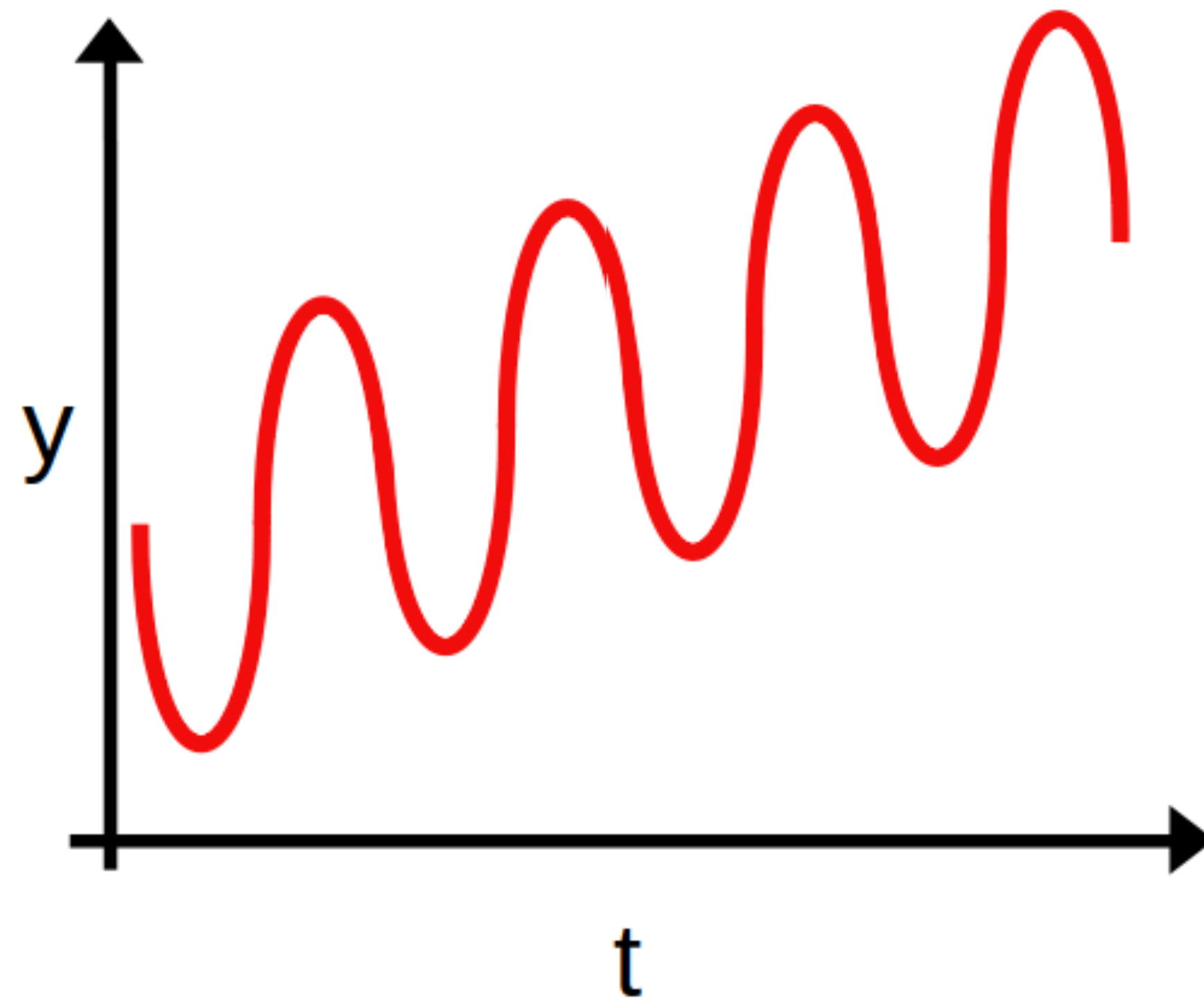


Implicit neural representation

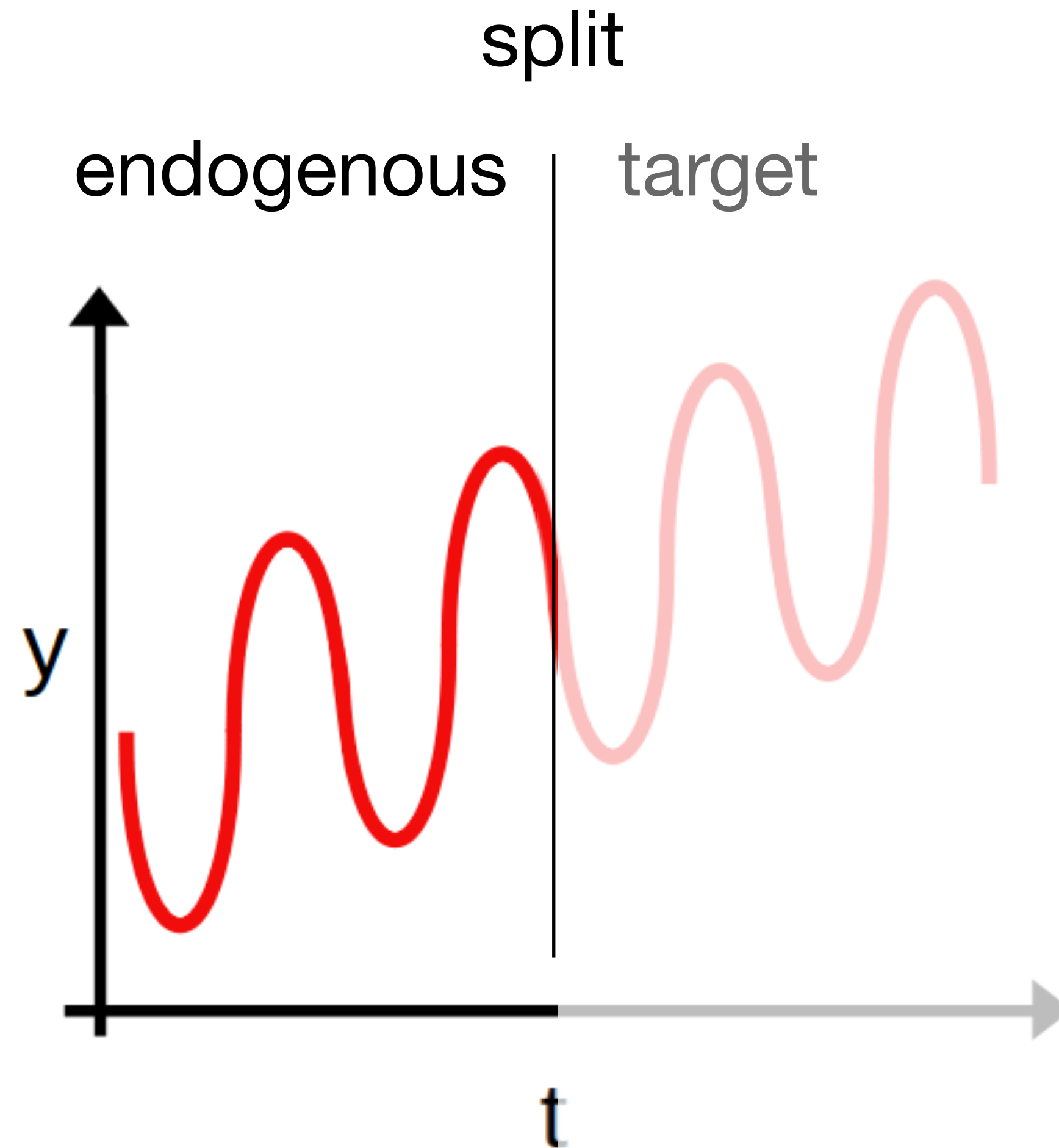


How can we modulate it ?

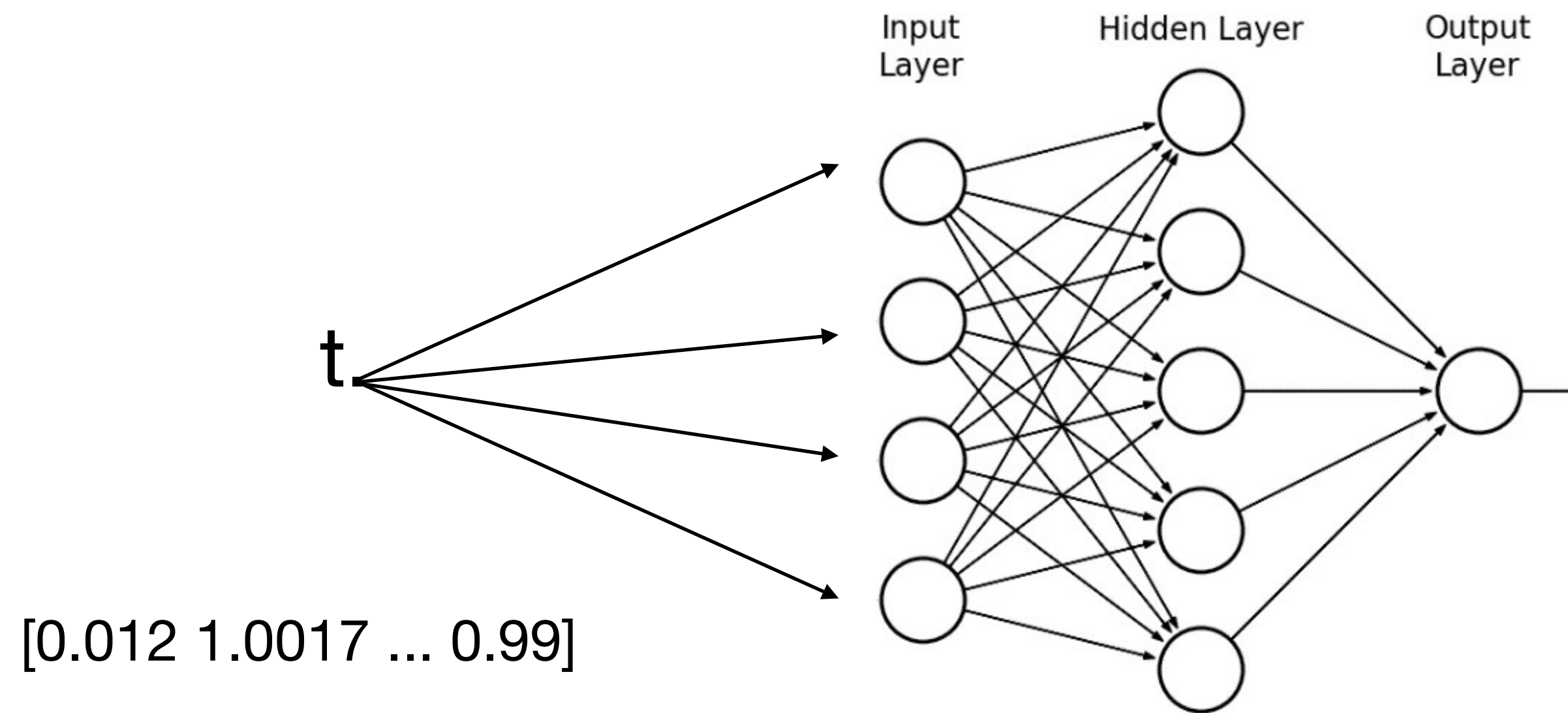
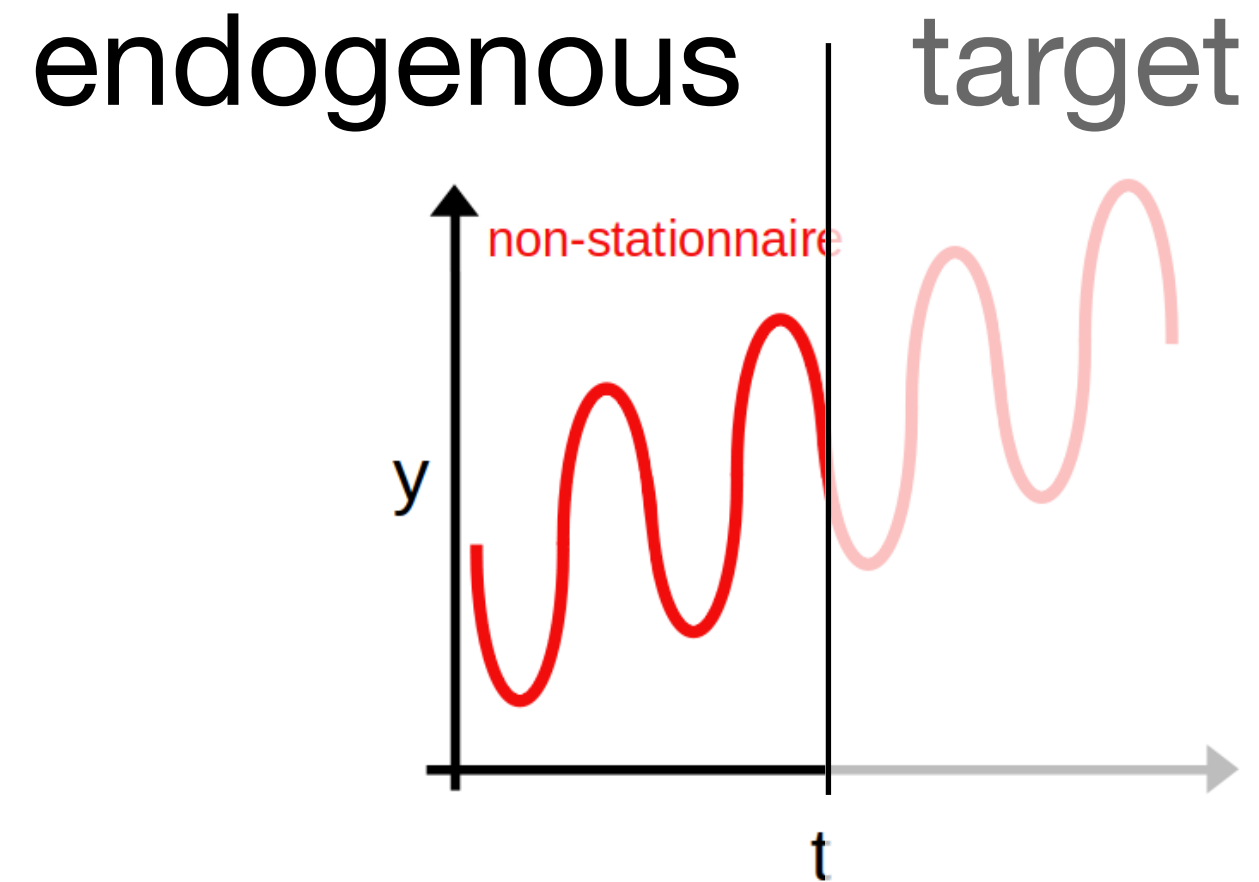
Modulate Implicit neural representation



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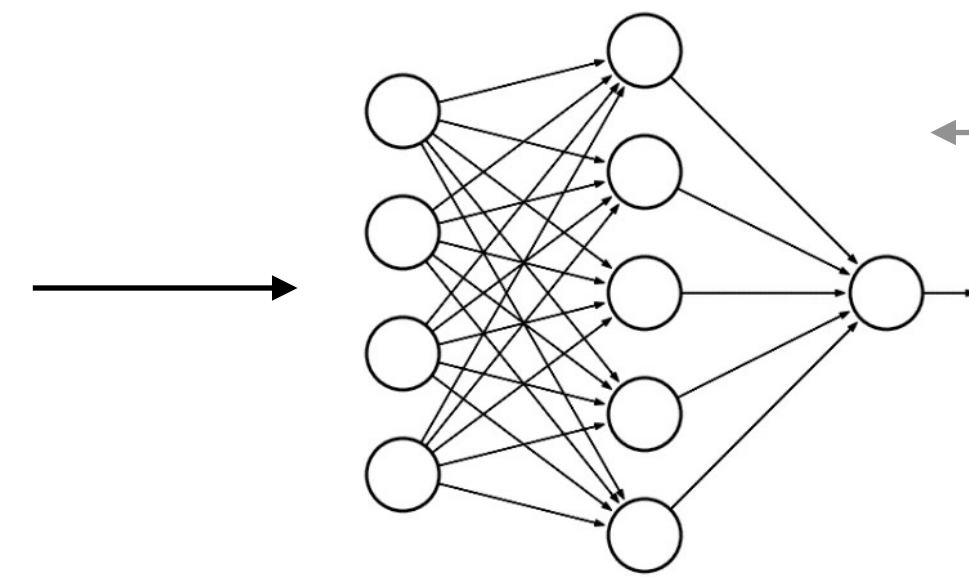
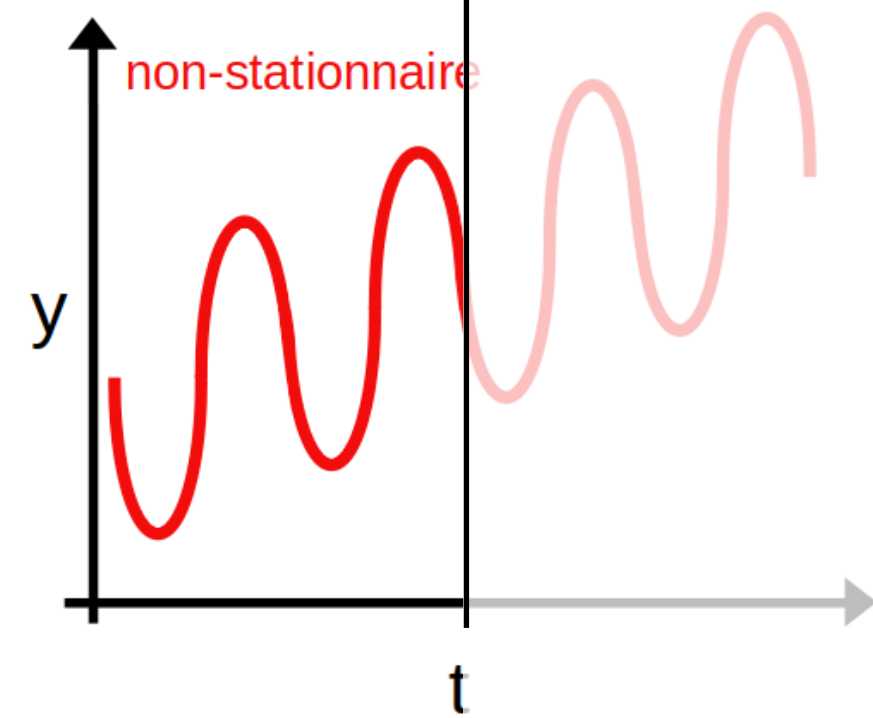


Modulate Implicit neural representation

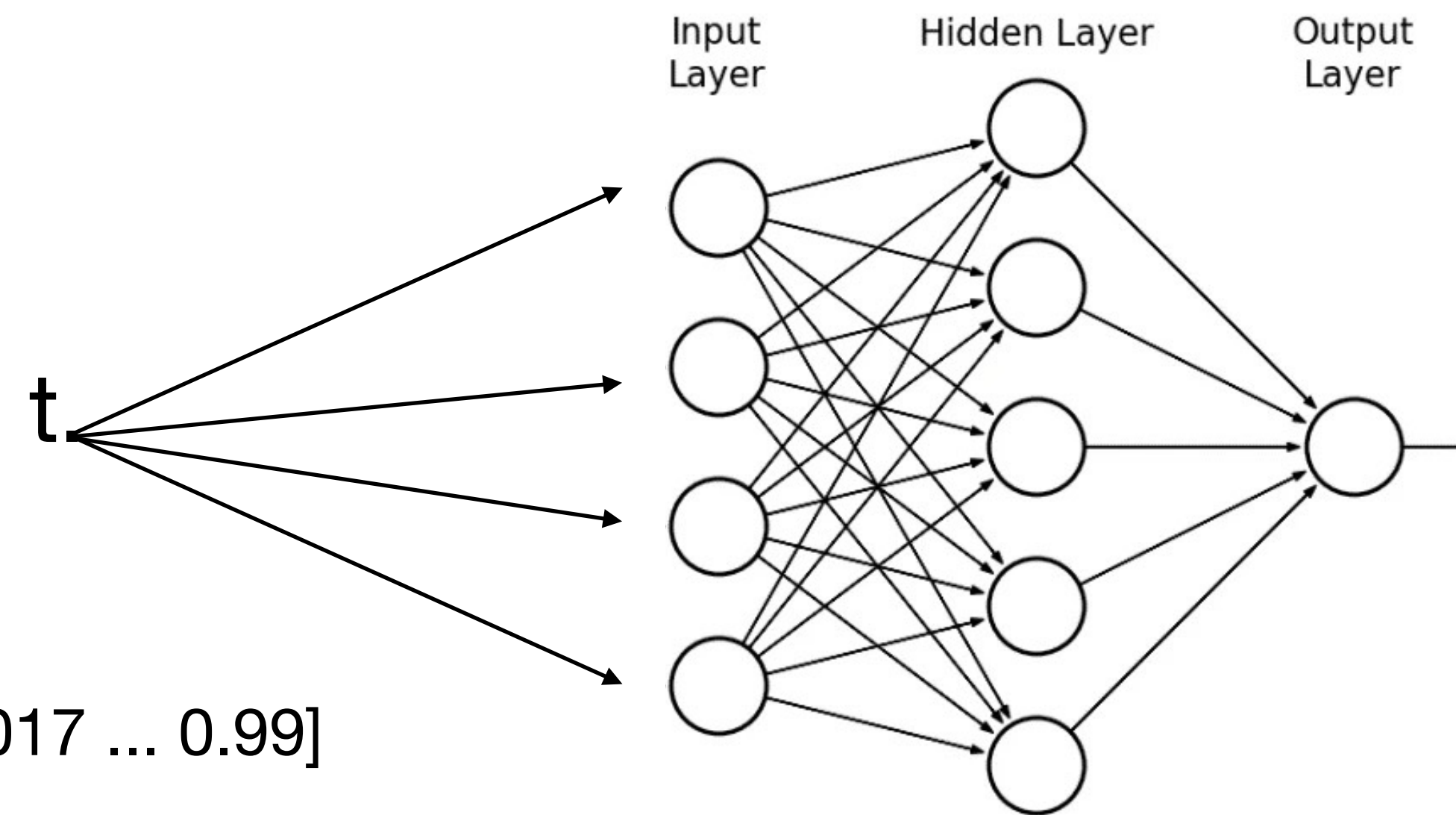


Modulate Implicit neural representation

endogenous target

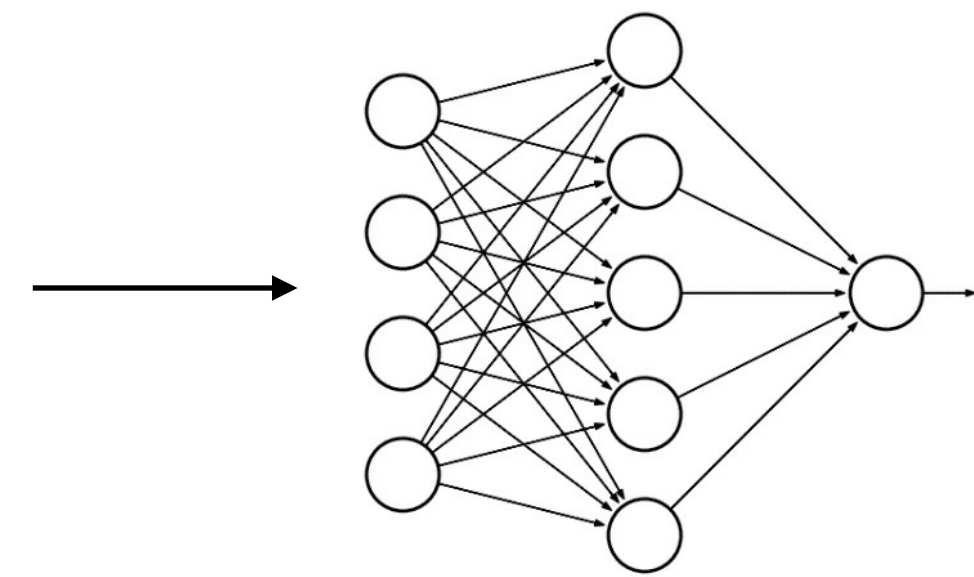
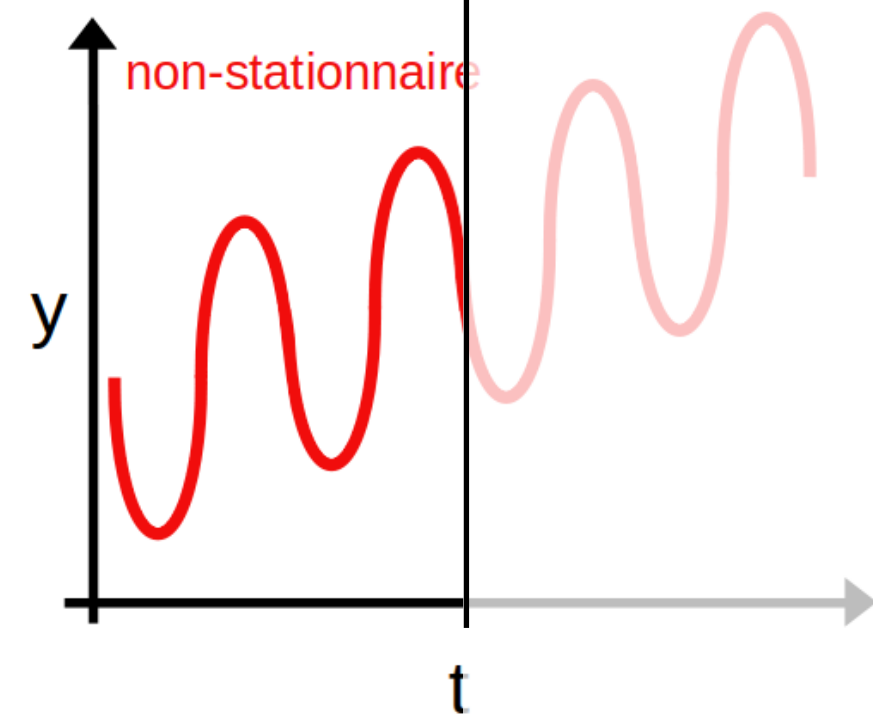


hyper network

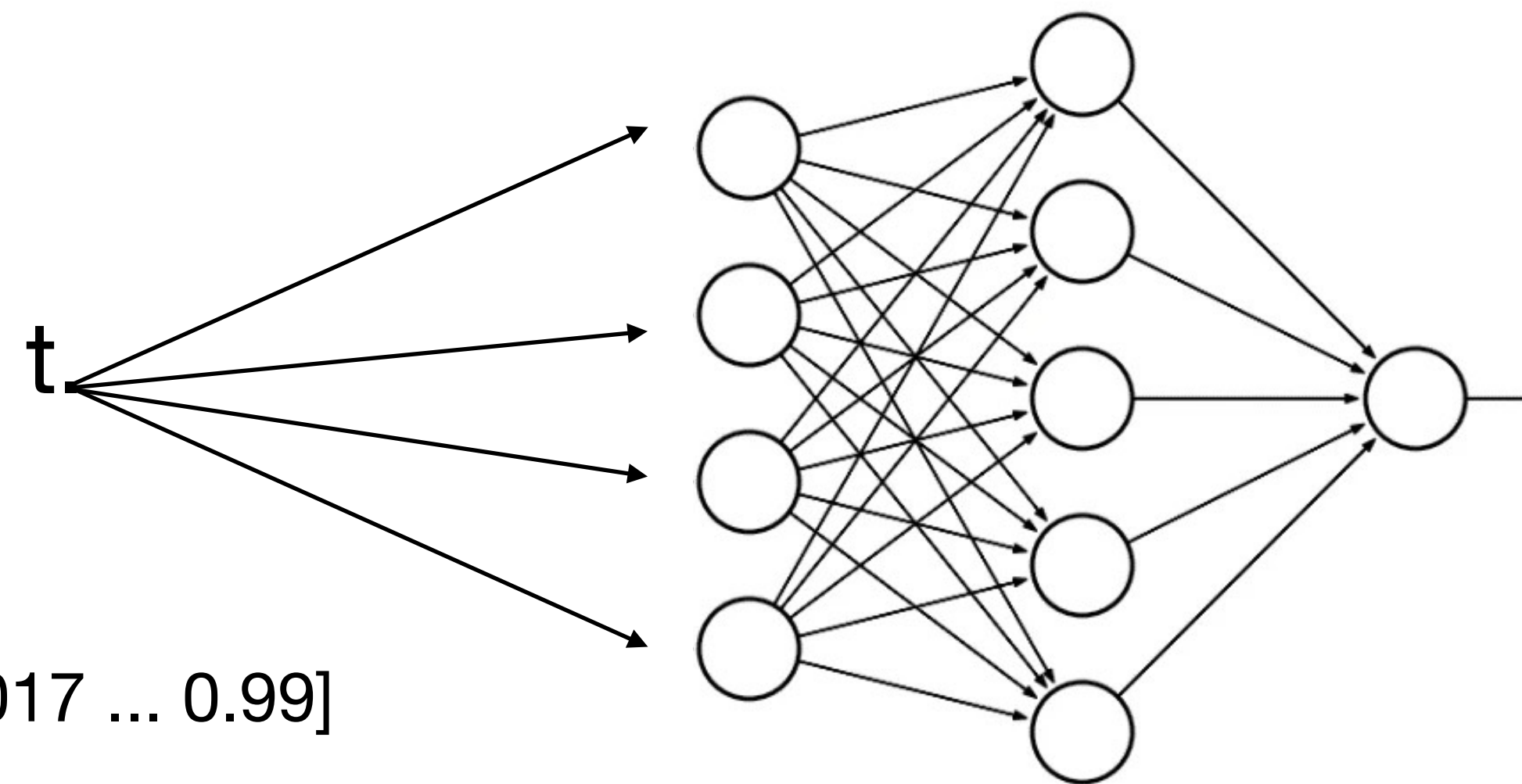


Modulate Implicit neural representation

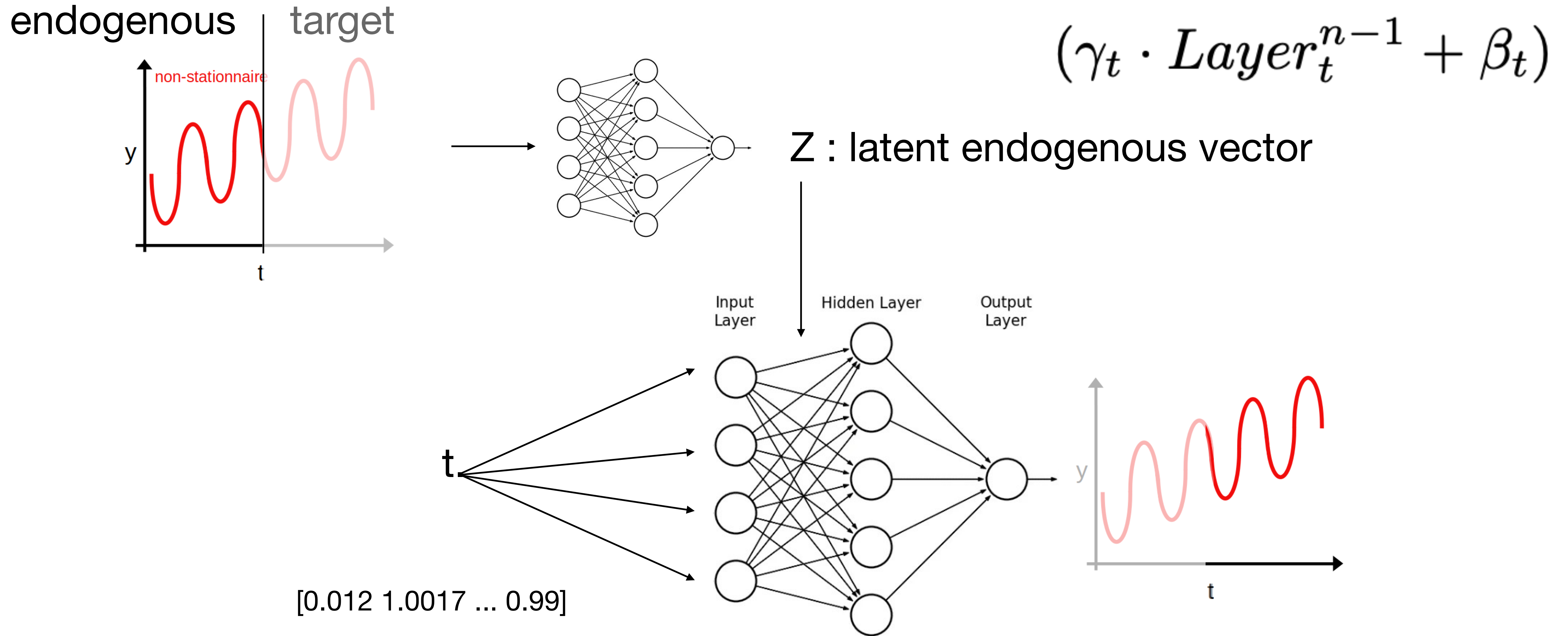
endogenous target



Z : latent endogenous vector



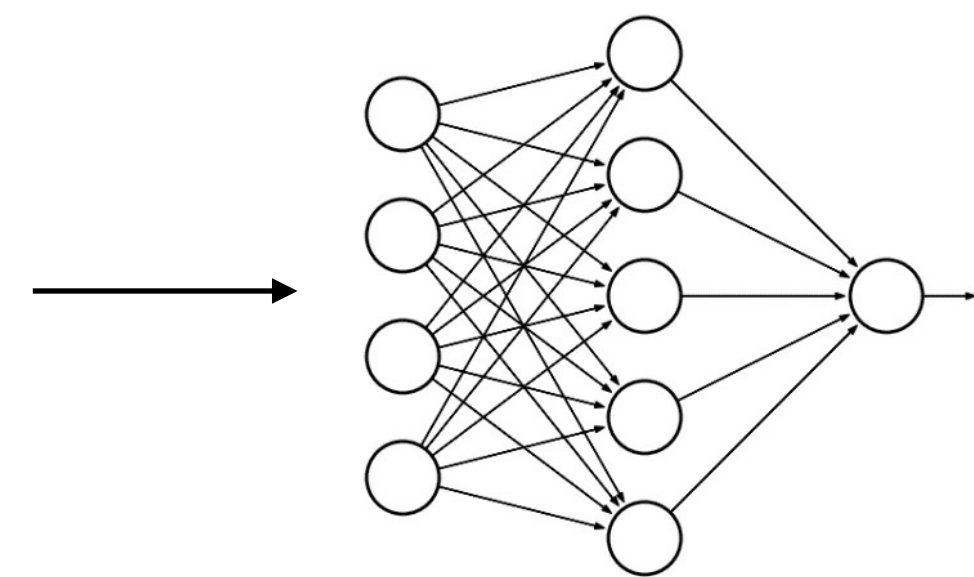
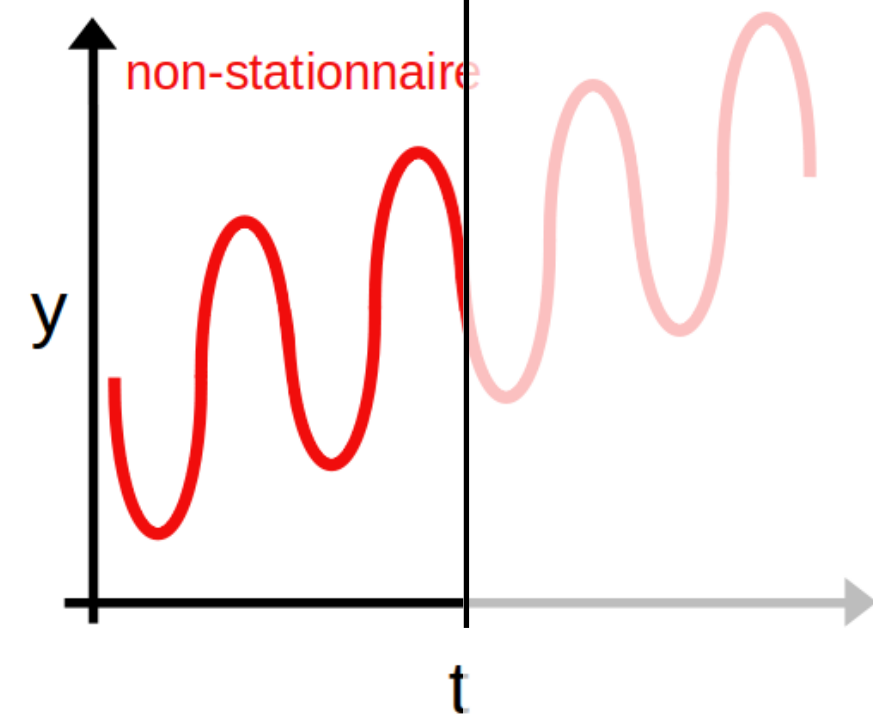
Modulate Implicit neural representation



Perez, E., Strub, F., de Vries, H., Dumoulin, V., & Courville, A. (2018). *FiLM: Visual Reasoning with a General Conditioning Layer*. Proceedings of the AAAI Conference on Artificial Intelligence, 32(1), 3942–3951.

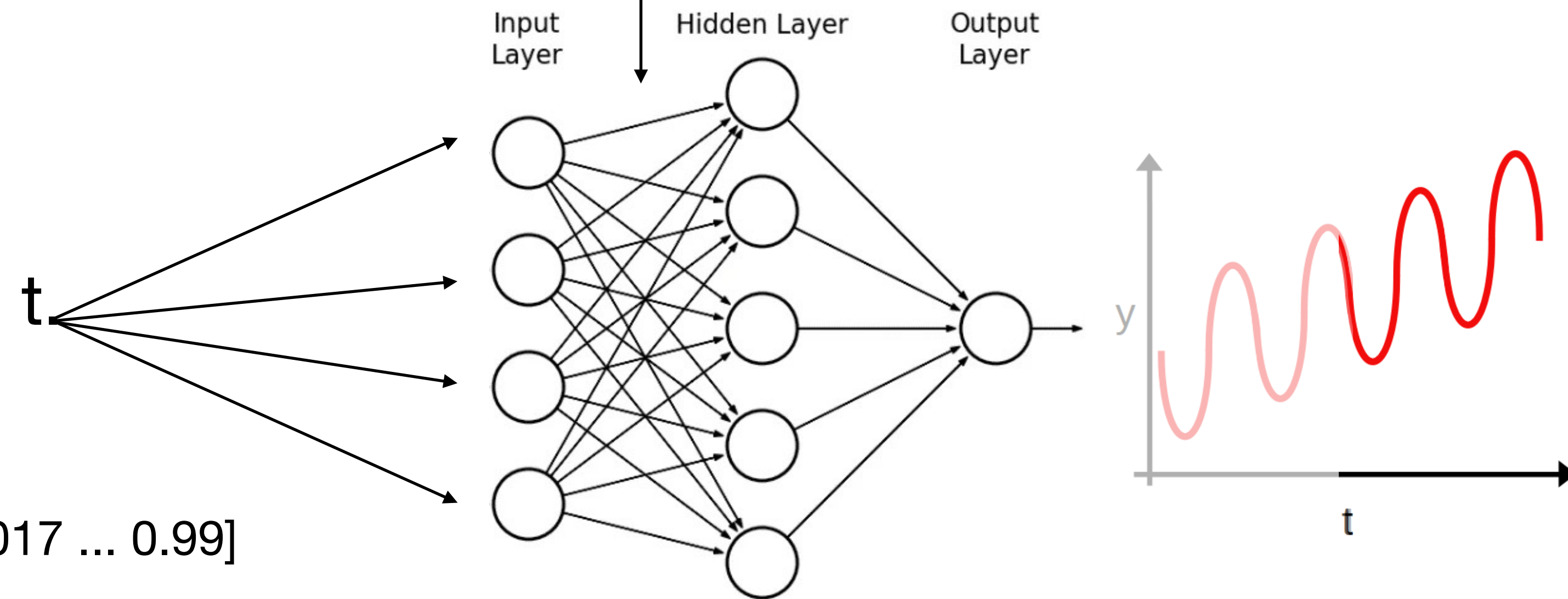
Modulate Implicit neural representation

endogenous target



Z : latent endogenous vector

$$(\gamma_t \cdot Layer_t^{n-1} + \beta_t)$$



Perez, E., Strub, F., de Vries, H., Dumoulin, V., & Courville, A. (2018). *FiLM: Visual Reasoning with a General Conditioning Layer*. Proceedings of the AAAI Conference on Artificial Intelligence, 32(1), 3942–3951.

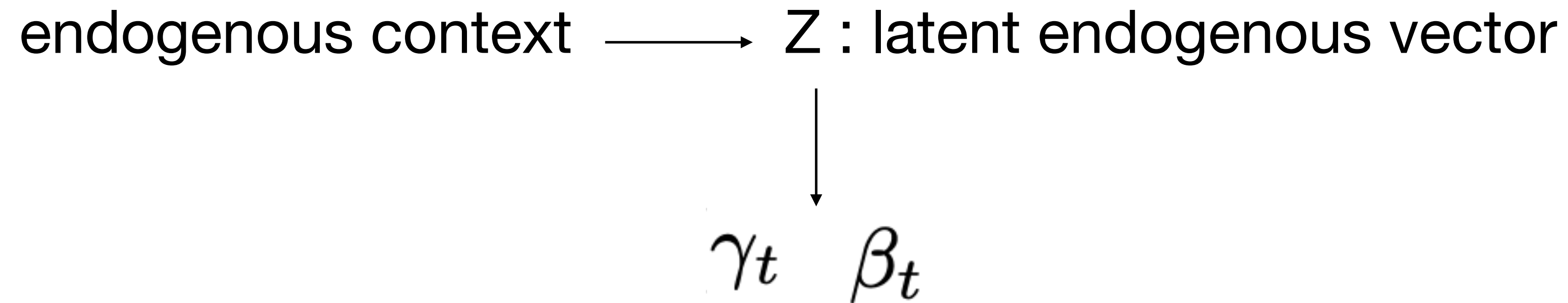
Modulate Implicit neural representation

endogenous context

Modulate Implicit neural representation

endogenous context $\longrightarrow$ Z : latent endogenous vector

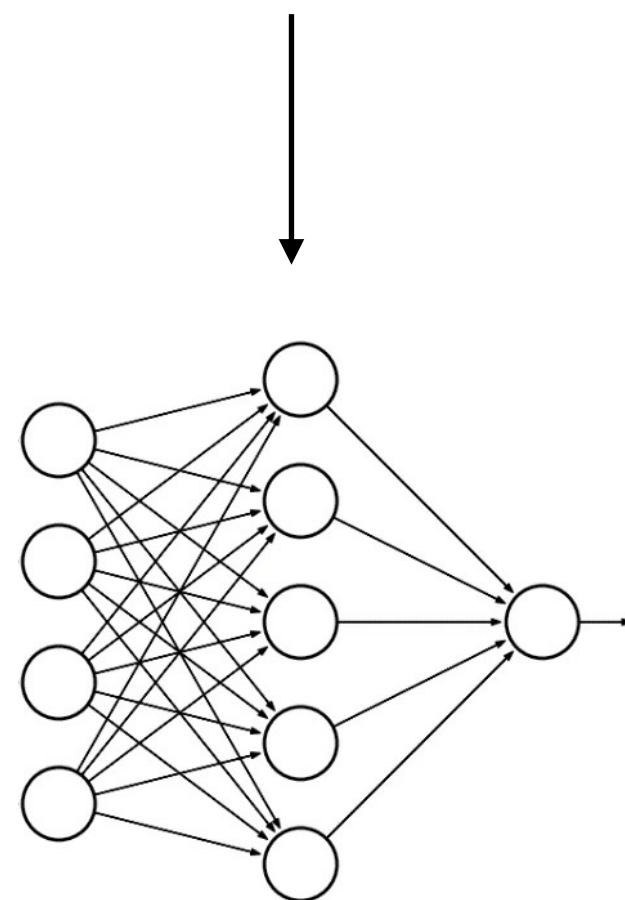
Modulate Implicit neural representation



Modulate Implicit neural representation

endogenous context $\longrightarrow$ Z : latent endogenous vector

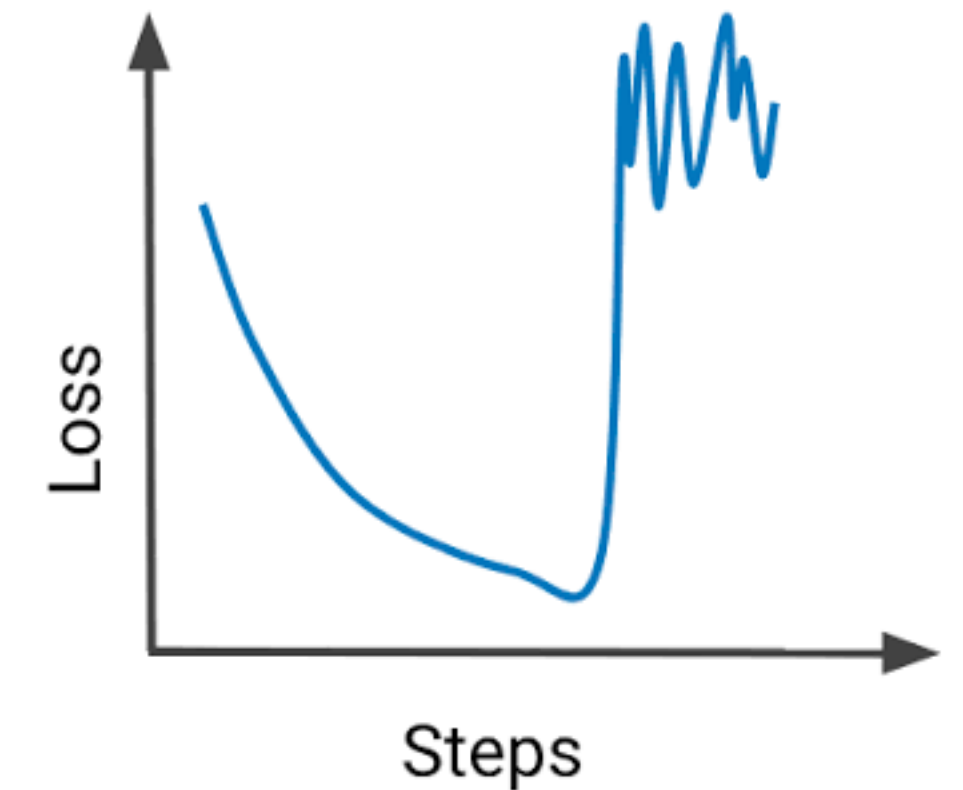
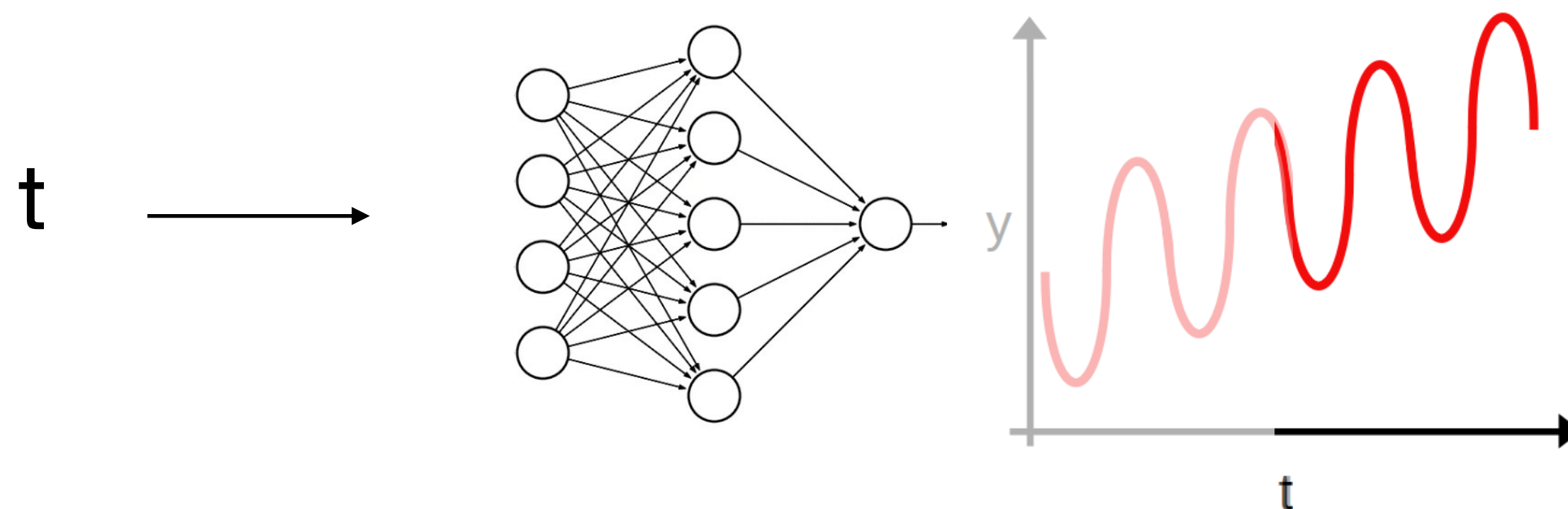
$$\gamma_t \quad \beta_t \quad (\gamma_t \cdot \text{Layer}_t^{n-1} + \beta_t)$$



Modulate Implicit neural representation

endogenous context $\longrightarrow$ Z : latent endogenous vector

$$\gamma_t \quad \beta_t \quad (\gamma_t \cdot \text{Layer}_t^{n-1} + \beta_t)$$



What we've done

Endogenous context VS Exogenous context

Endogenous context = information from the target itself

Exogenous context = information from outside the target

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How can we leverage dynamic exogenous context for time series forecasting?

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How can we leverage dynamic exogenous context for time series forecasting?

Concatenate in hypernetwork !

What we've done

Endogenous context VS Exogenous context

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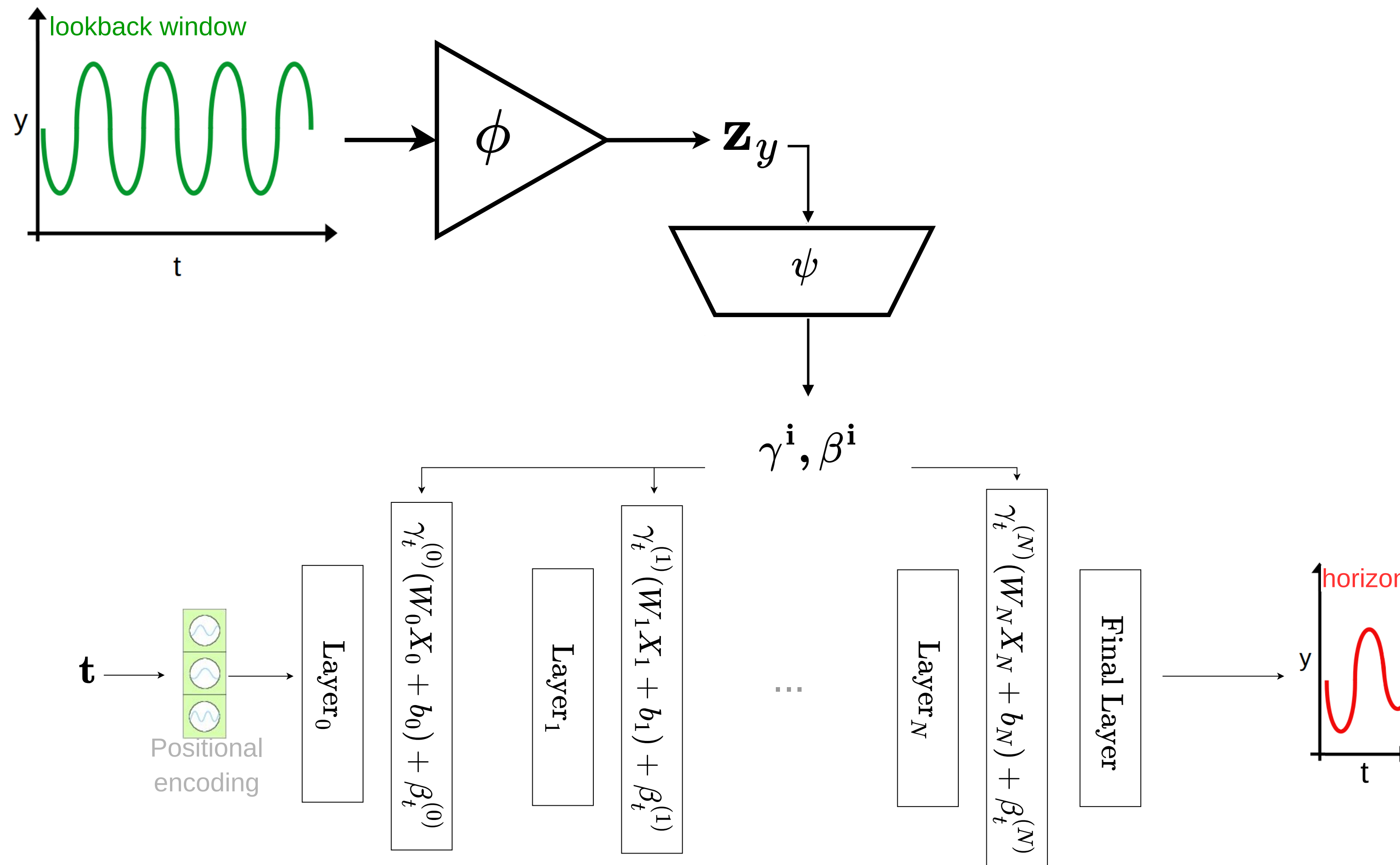
Exogenous context = information from outside the target

How can we leverage dynamic exogenous context for time series forecasting?

Concatenate in hypernetwork !

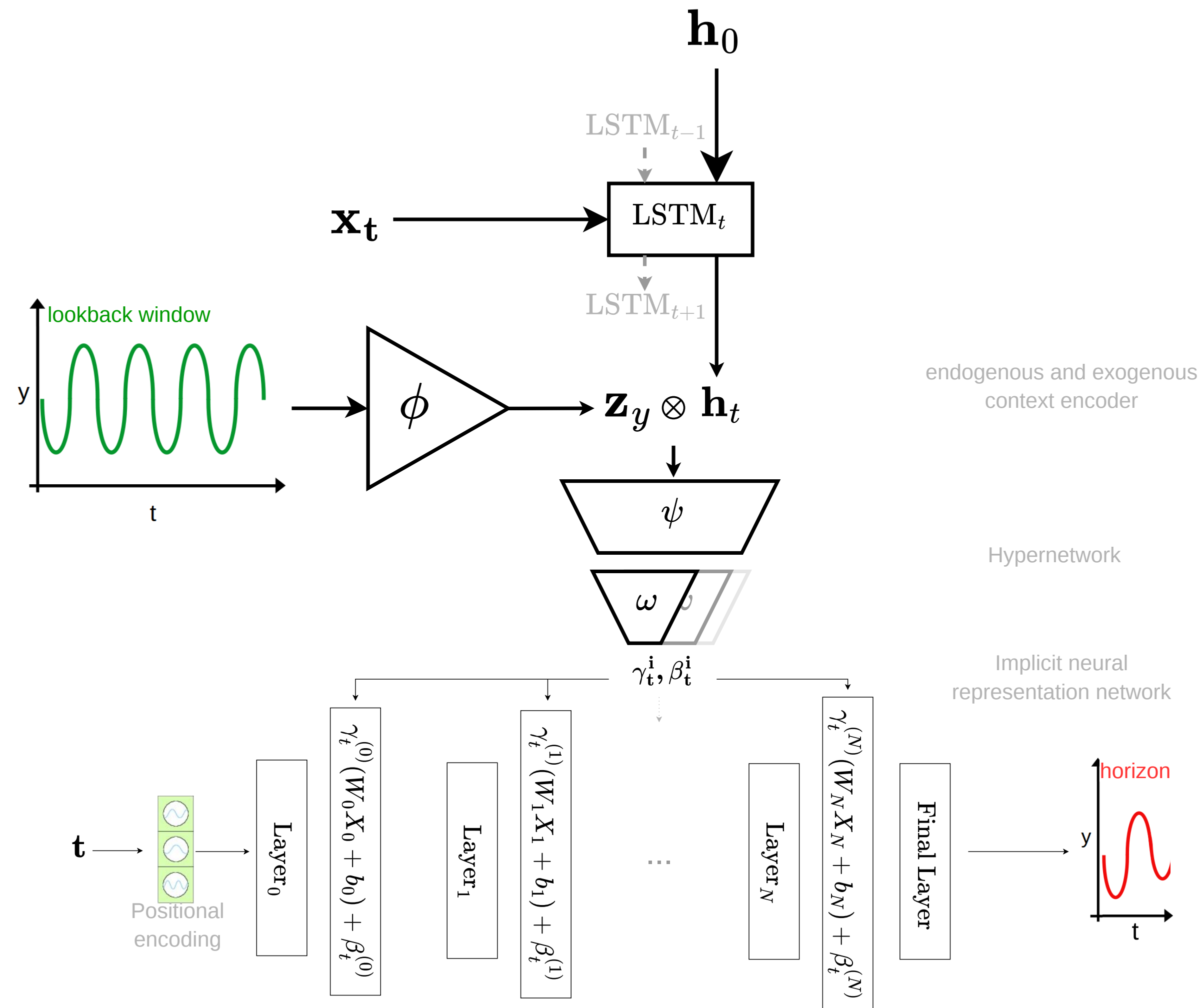
How should models adapt when the **context changes over time?**

Modulate Implicit neural representation

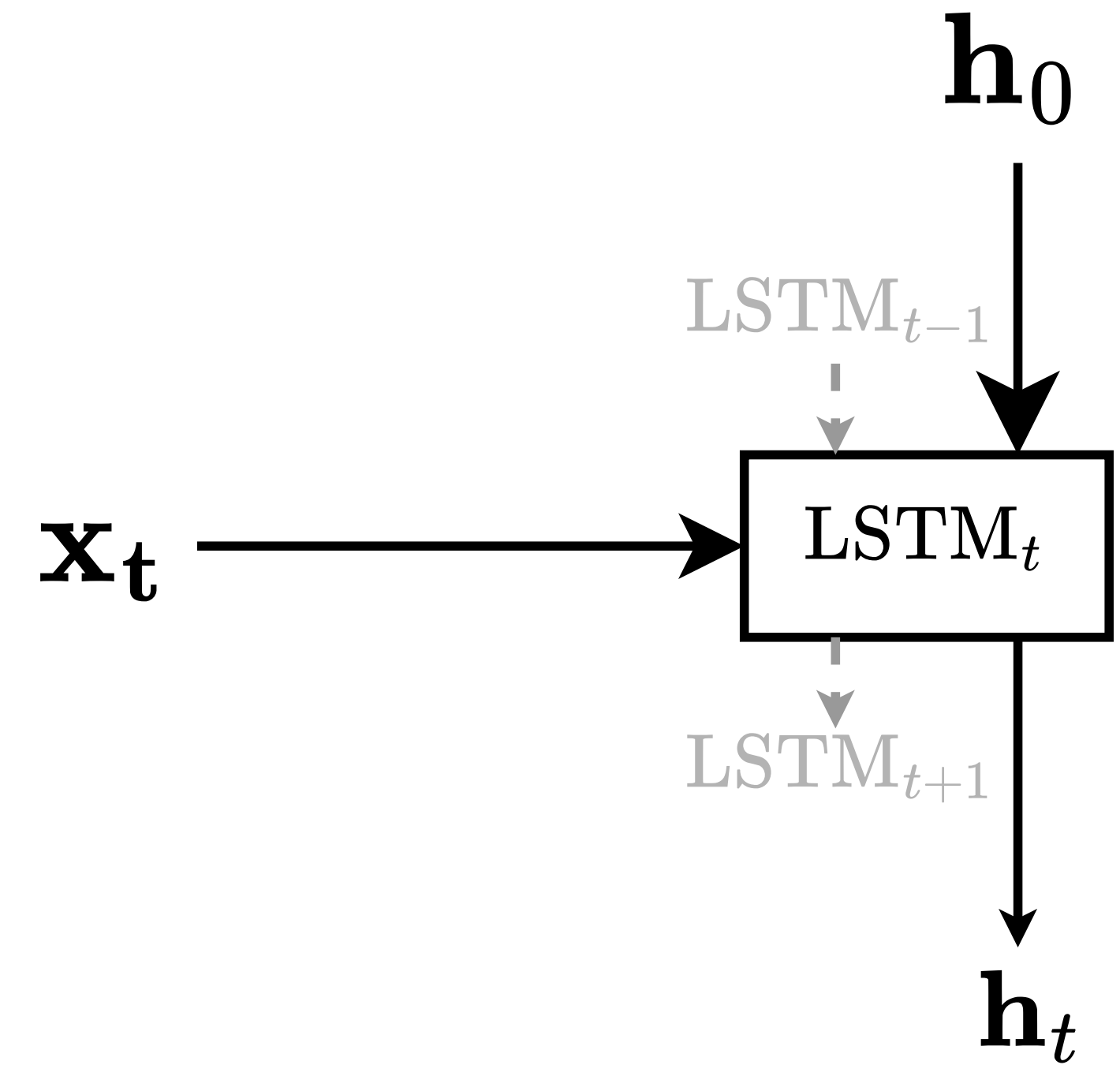
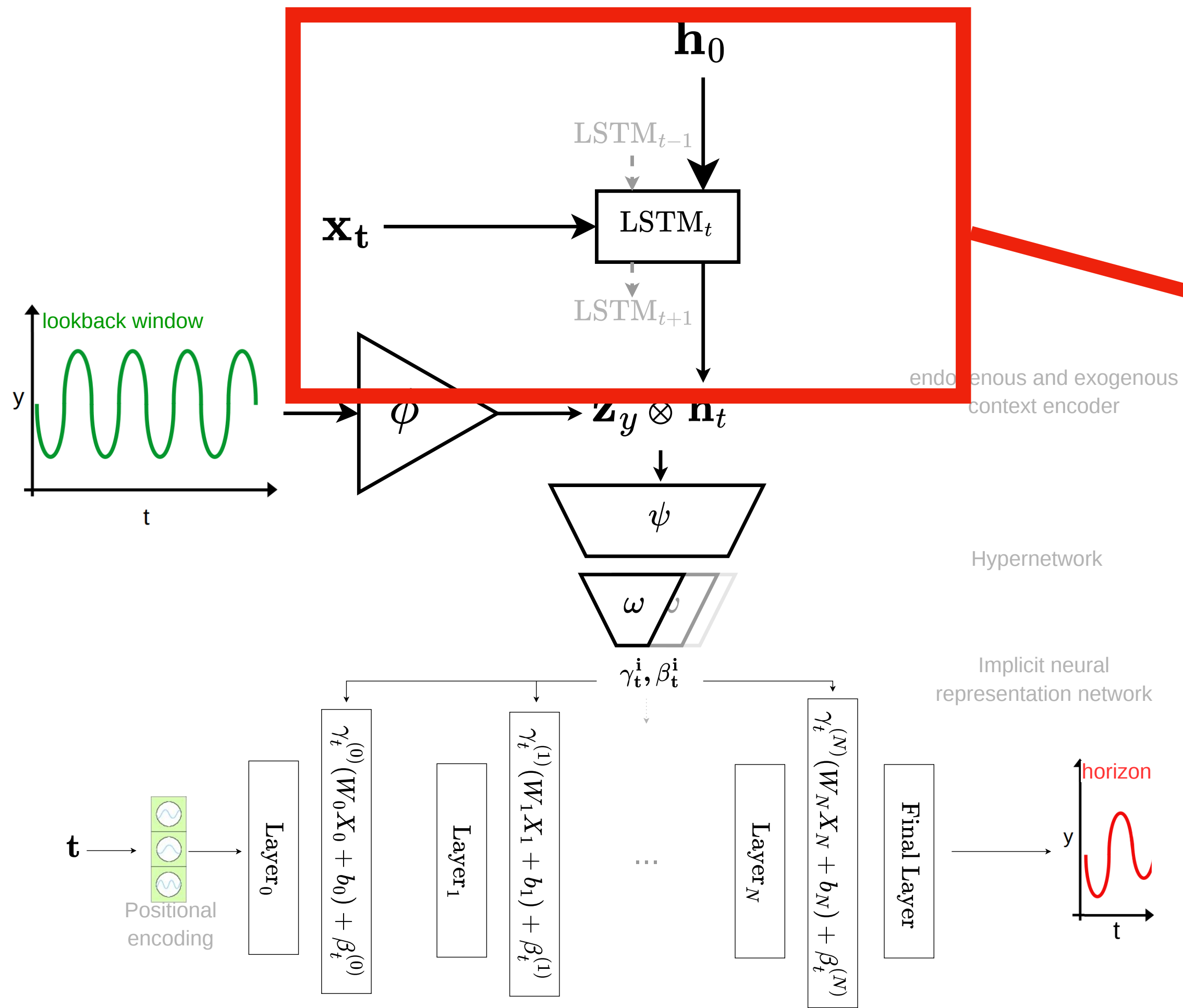


Le Naour et al., "Time Series Continuous Modeling for Imputation and Forecasting with Implicit Neural Representations," TMLR, 2024.

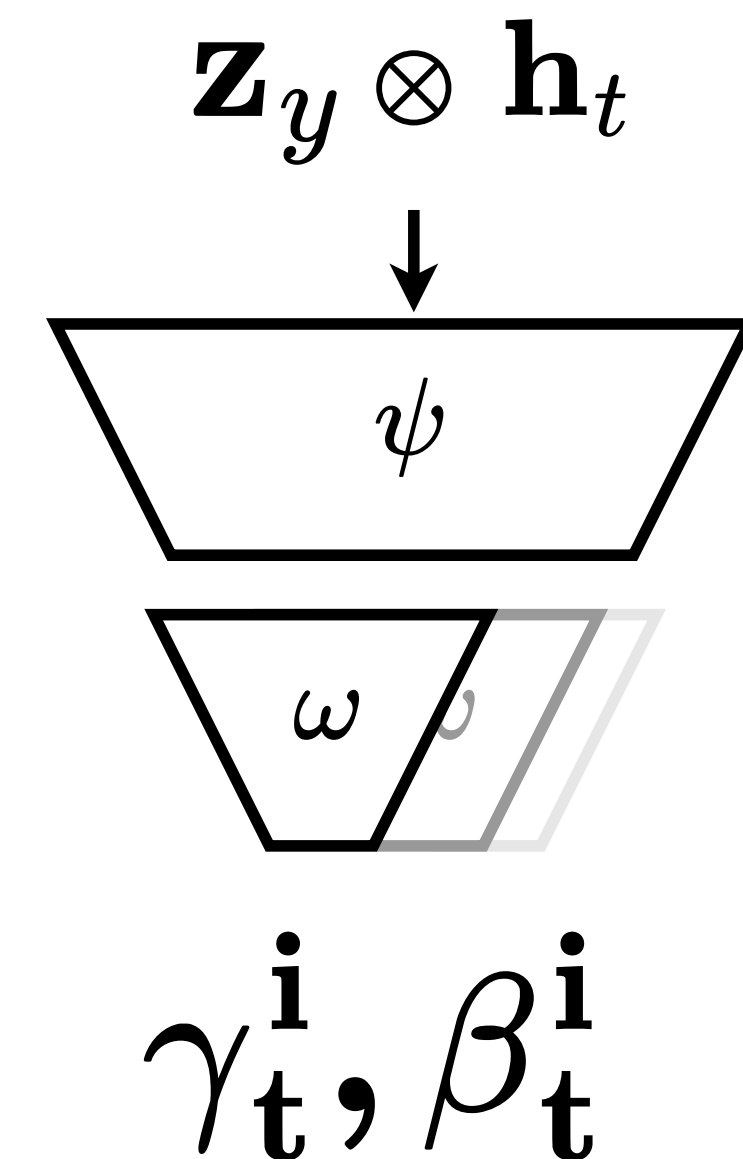
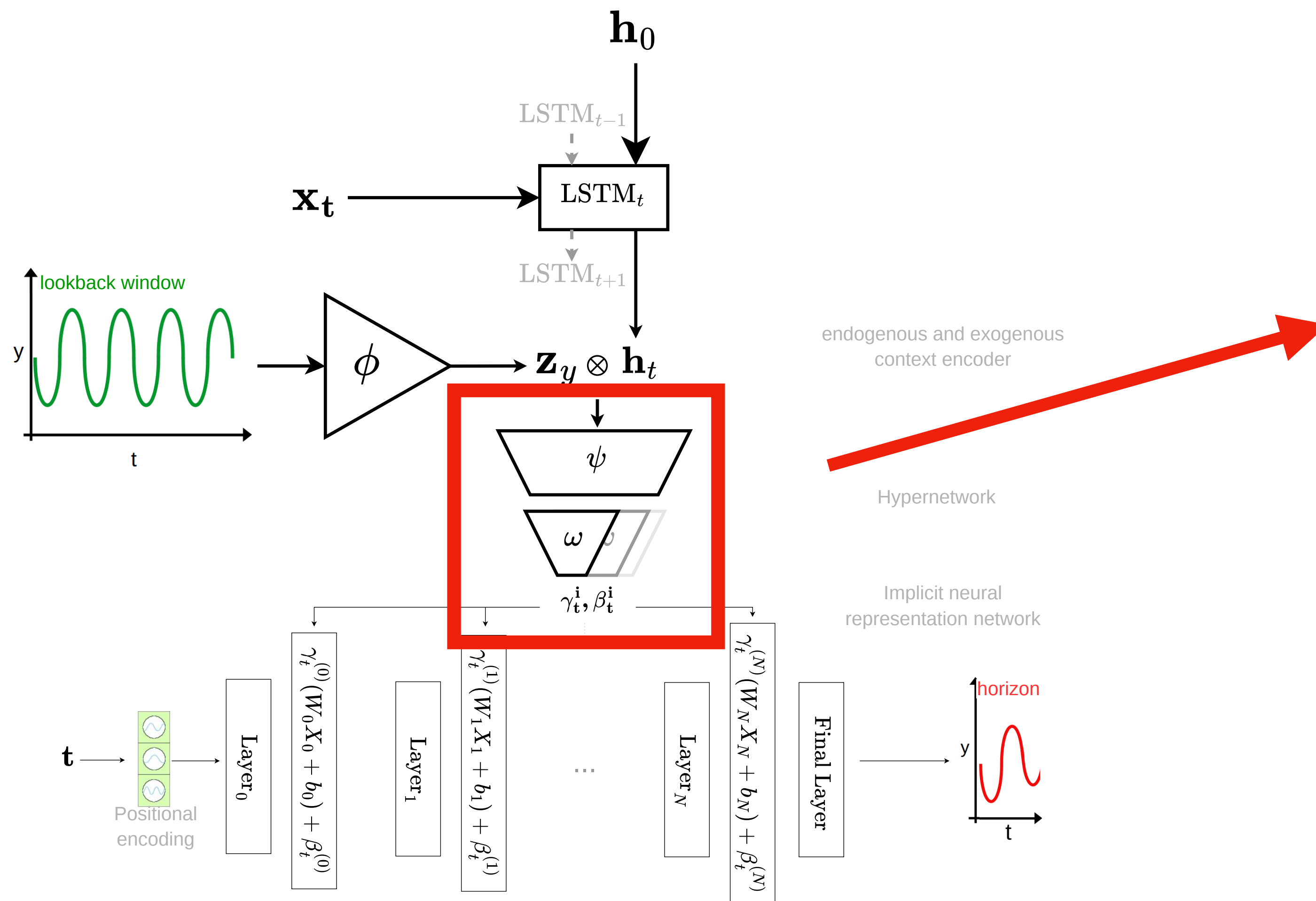
What we've done



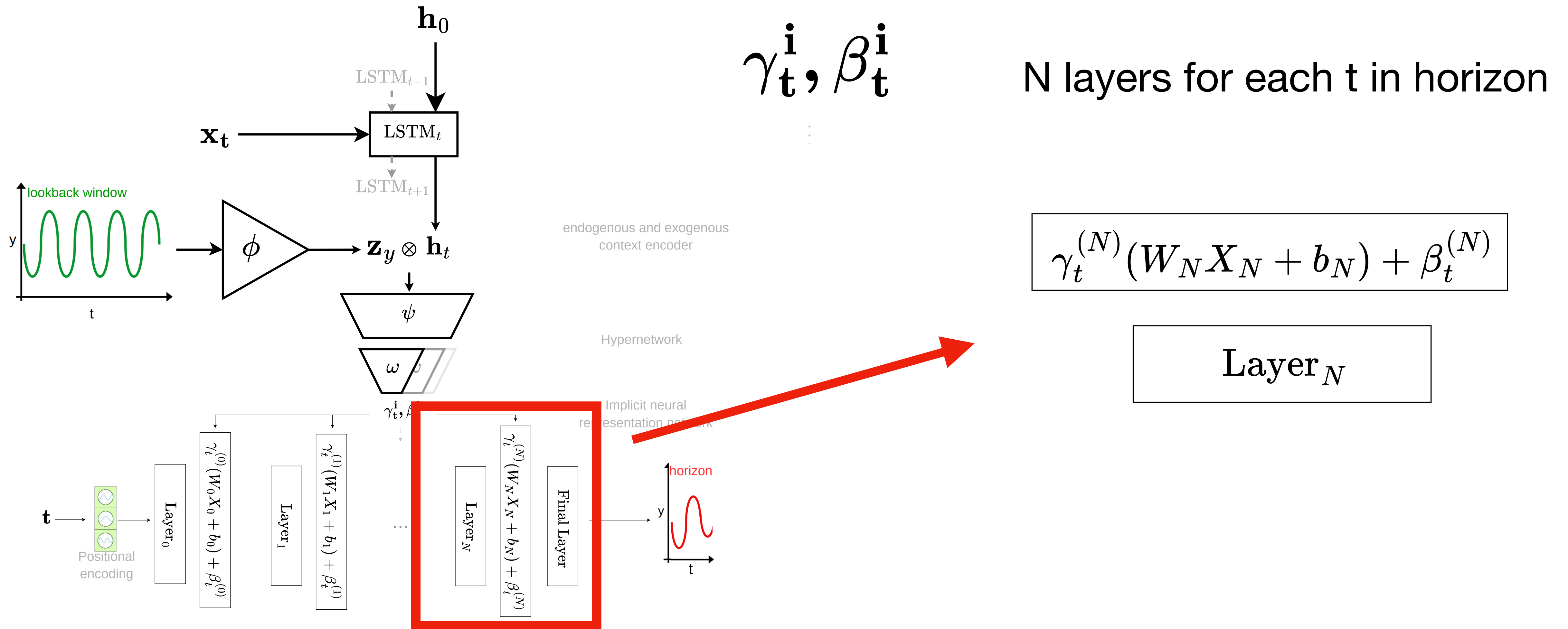
What we've done



What we've done



What we've done

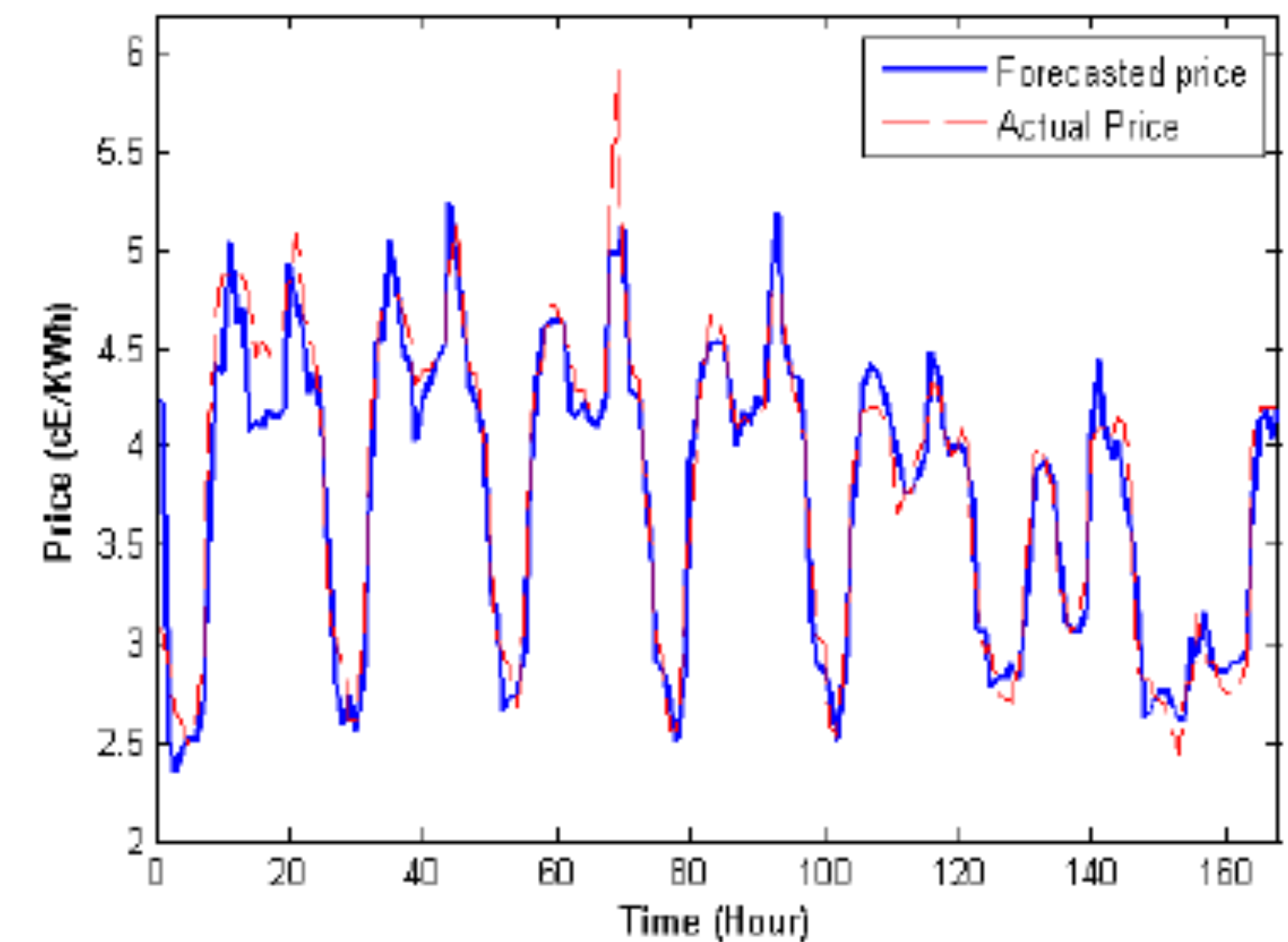


Electricity price forecasting

EPF Benchmark

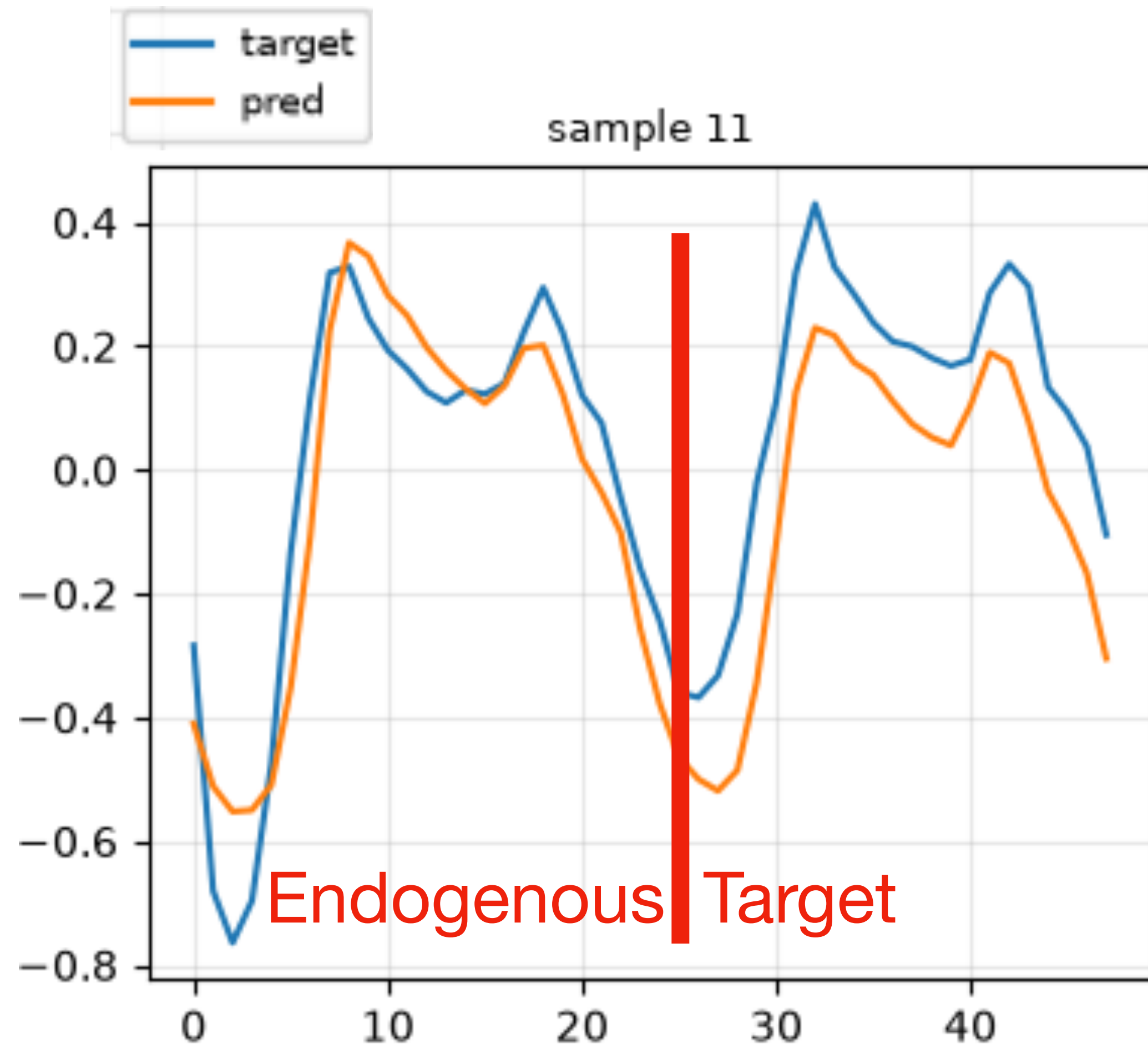
Electricity Price Forecasting

- ⚡ **Task:** day-ahead electricity price forecasting
- 🕒 **Hourly** time series
- 🌐 **Multiple European electricity markets**
- 📊 **Input:** historical prices + exogenous variables
- 🌟 **Horizon:** next **24 hours**



Lago et al., *Forecasting day-ahead electricity prices: A review of state-of-the-art algorithms, best practices and an open-access benchmark*, **Applied Energy**, 2021.

Electricity price forecasting



Exogenous context:

- ⚡ Electricity demand (Load forecast)
- 🌬️ Wind power generation (forecast)

Results

Table 1. Effect of removing the exogenous dynamic context (mean $\pm$ standard deviation, $n = 5$ seeds, EUR/MWh)

Model	Context	MSE	RMSE	MAE	SMAPE (%)	p -value (MSE)
INR	with	24.78 $\pm$ 4.74	4.96 $\pm$ 0.47	3.05 $\pm$ 0.52	7.89 $\pm$ 1.28	0.0141
	without	31.63 $\pm$ 1.13	5.62 $\pm$ 0.10	3.41 $\pm$ 0.10	8.73 $\pm$ 0.28	
LSTM	with	32.74 $\pm$ 1.21	5.72 $\pm$ 0.11	3.67 $\pm$ 0.11	9.22 $\pm$ 0.29	0.0417
	without	36.70 $\pm$ 2.01	6.06 $\pm$ 0.16	3.74 $\pm$ 0.33	9.65 $\pm$ 0.77	
Average baseline	n/a	40.71	6.38	3.96	10.29	—

Note: p -value of the paired t -test ($n = 5$ seeds) comparing configurations with vs. without dynamic context, calculated based on the MSE.

Conclusion

–22%: INR error with context (31.63 → 24.78, $p = 0.014$)

–11%: LSTM error with context (36.70 → 32.74, $p = 0.042$)

Exogenous dynamic context is a key driver of performance, especially for INR

Limits to keep in mind

Higher variance for INR with context.

Single window size

Future work

Window size sensitivity

Try other datasets.

Long-term horizon

This is the end

Thank for your attention !

Any question now or jessy.azizi@agroparistech.fr

Special thanks to you know who you are